Recent linguistic experience modulates abstraction of event roles

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How do people perceive and understand events? Prior research suggests that this hinges on people's extraction of information on the roles that entities in an event take, such as agent, patient, or instrument.[1-2]People can extract this type of information from real world scenes with only limited exposure to event scenes.[3-5] The question is, what factors underlie the efficiency with which people are able to apprehend events? Knowing that scene and event perception requires an interplay between bottom-up (stimulus-related perceptual features) and top-down factors,[6] here, we study the extent to which *language*-related factors play a role in guiding event apprehension, in a top-down fashion. More specifically, can recent linguistic experience influence people's in-the-moment extraction of event roles from real-world event stimuli? Prior work has shown a unique role for linguistic labels during perceptual processing: people are presented with linguistic (words) or nonlinguistic information (environmental sounds) ('cues') shortly prior to visual stimuli. Linguistic cues consistently lead to an advantage in people's performance on perceptual detection and discrimination tasks.[7] Here, we study a potential linguistic cueing advantage for visual perception beyond single objects: We examine whether verbs with different syntactic frames (transitive, e.g., to hug, or intransitive verbs, e.g., to cry) may influence the processing of subsequently viewed *events* (e.g., boy hugging girl). We expect (Turkish) verbs to evoke abstract event representations that involve either one or two event participants. Specifically, transitive verbs—whose argument structure licenses two event participants in agent–patient roles—should facilitate the processing of visually presented Agent–Patient events, and intransitive verbs—associated with single-participant (Agent-only) structures—should impede this process, as an event template with only 1 participant is activated. Turkish participants are briefly exposed to human Agent–Patient scenes (for 300 ms), preceded by transitive verbs, intransitive verbs, or no-verb controls (nonsense word). Overlap between a transitive verb's event structure and the visual event should support balanced attention to both event roles, whereas incongruence introduced by intransitive verbs will selectively reduce expectations regarding the patient role relative to the control condition. By contrasting verbs that share syntactic frames but differ in event meaning (e.g., *hug* vs. *pull* preceding a hugging event), Exp. 2 also addresses whether argument structure alone, or the specific semantic relation linking participants, drives these effects. Event role identification is measured by studying accuracy and RTs in response to probe pictures (with Role matching, or Role mismatching set-ups) presented immediately after the briefly presented stimuli ("was the second picture the one you saw earlier?"). Findings (Exp. 1) show a facilitating effect of Transitive verbs on reaction times in Role match trials: A cued event structure with two participants facilitates the processing of events with similar event structure (Fig. 1). Exp.2, however, shows no significant effects of Prime type (Fig. 2). An abstract event structure, mismatching in semantics with the probe events, does not cue event processing in the same way. Overall, findings show how verb meaning and syntactic structure can jointly lead to an in-the-moment top-down modulation of event role extraction, but not syntactic structure in an abstract form, such as an event structure encoded in a verb's argument structure, alone.

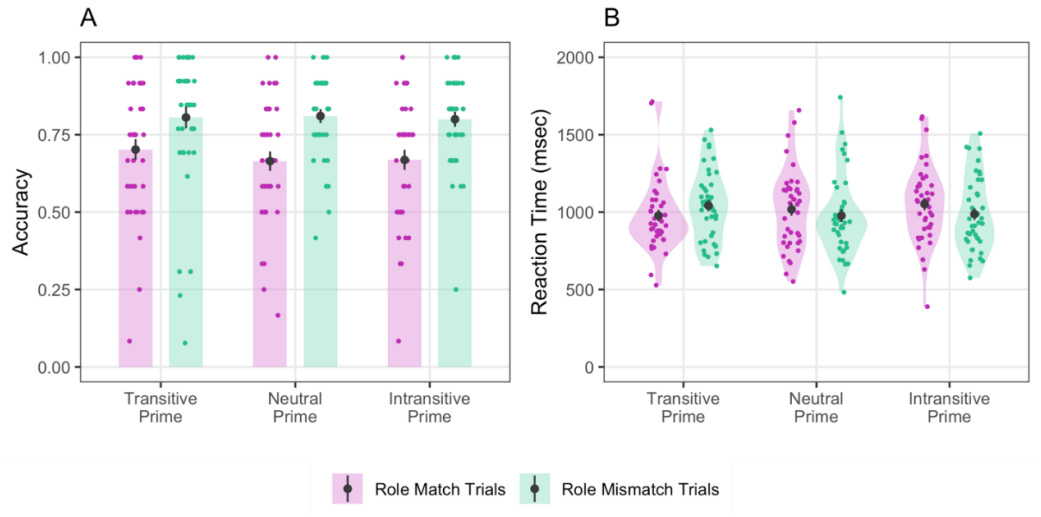

Fig. 1: Accuracy (A), RT (B) in the probe-recognition task in **Exp. 1** (Transitive primes **match** event semantics of probe pictures). Accuracy: ME of Picture type; RTs: interaction Picture type * Prime condition.

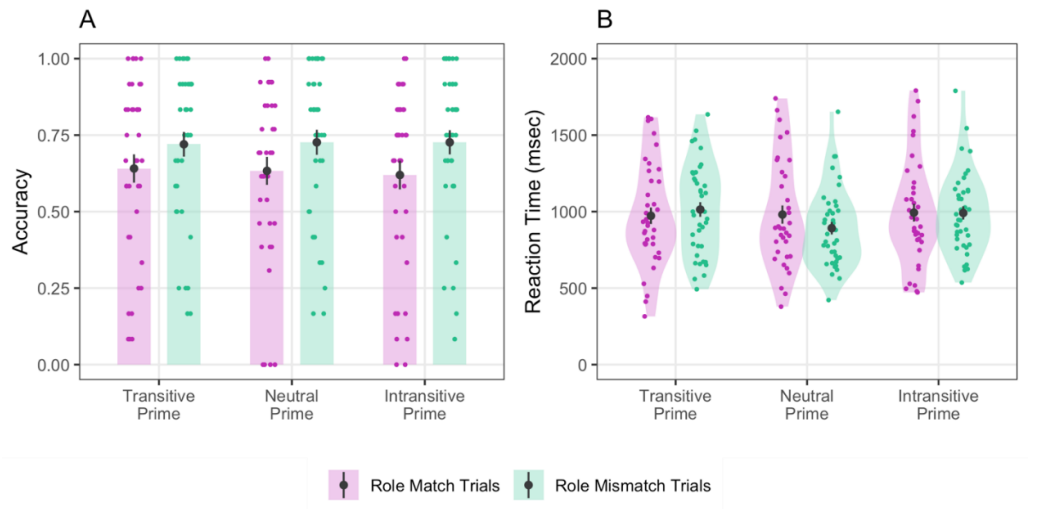

Fig. 2: Accuracy (A), RT (B) in the probe-recognition task in **Exp. 2** (Transitive primes do **not** match event semantics of probe pictures). Accuracy: ME of Picture type; RTs: n.s.

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